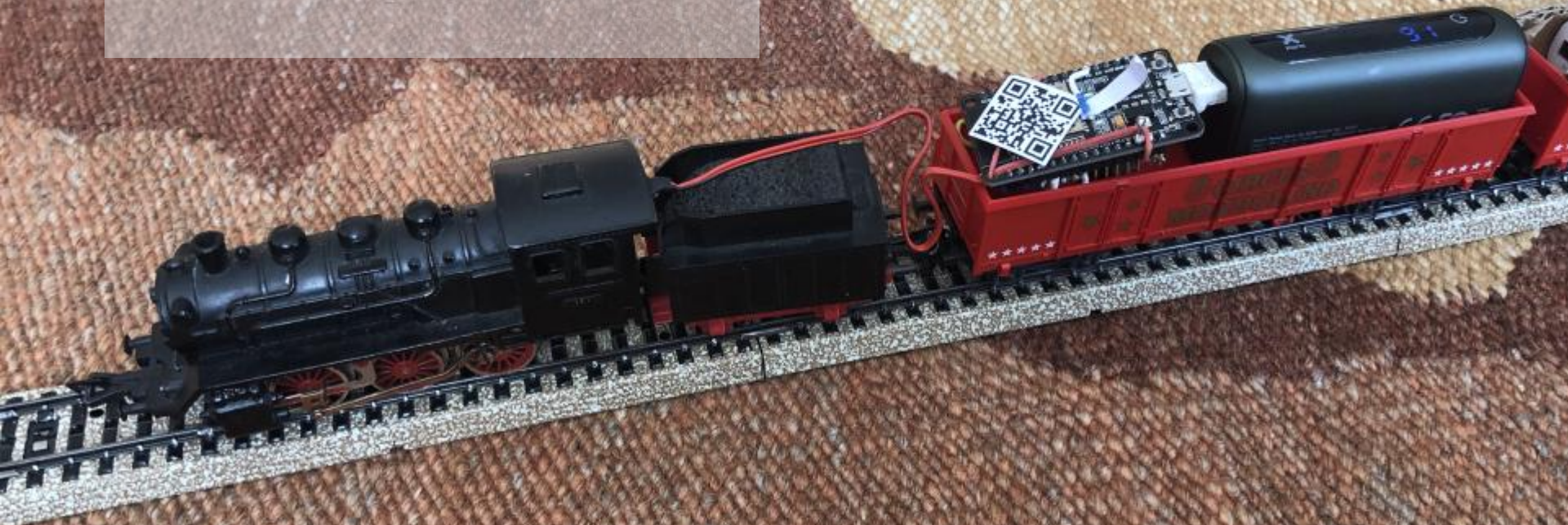


PiedPiperS

smartphone speed control
for model trains



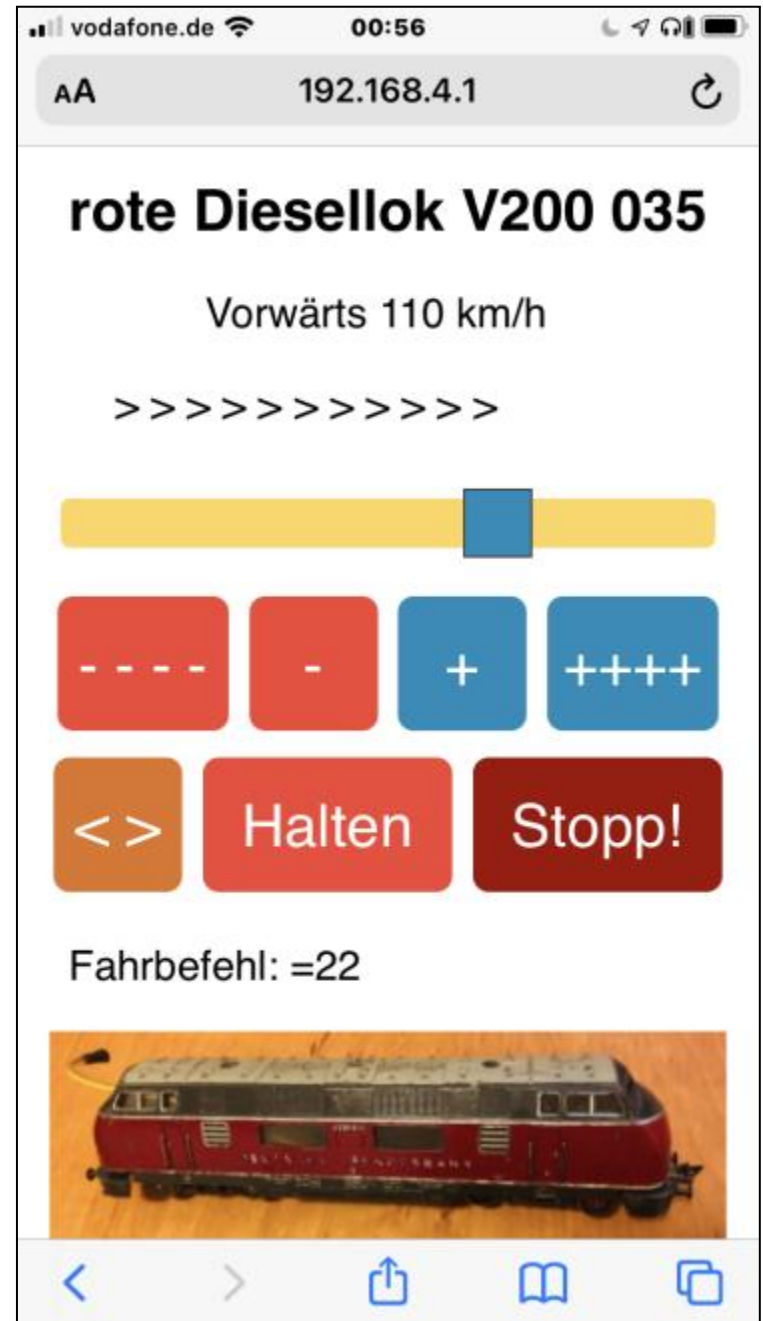
- Outdoor compatible, on-board USB power supply independent of track current
- ESP32 microprocessor, C++ Arduino software, with WIFI control, access point and webserver
- No proprietary elements, construction time 1 weekend, material cost ca. 25 €

PiedPiperS outdoor model train operation



Train control website on smartphone

- Compatible with every smartphone with WIFI
- ESP32 WIFI access point mounted on train
- Browser access to control panel via webserver
- Stable WIFI remote control for up to two users on HTML website with addition of simple JavaScript functions
- Additional manual control via touch pin
- Feedback from train operation on two LEDs
- Motor IC error handling, short cut protection
- ESP32 voltage spike and brownout protection
- Option for online power measurement and data display in tables and charts
- Option for true speed measurement by IR sensor detection of railway sleepers



PiedPiperS touch commands

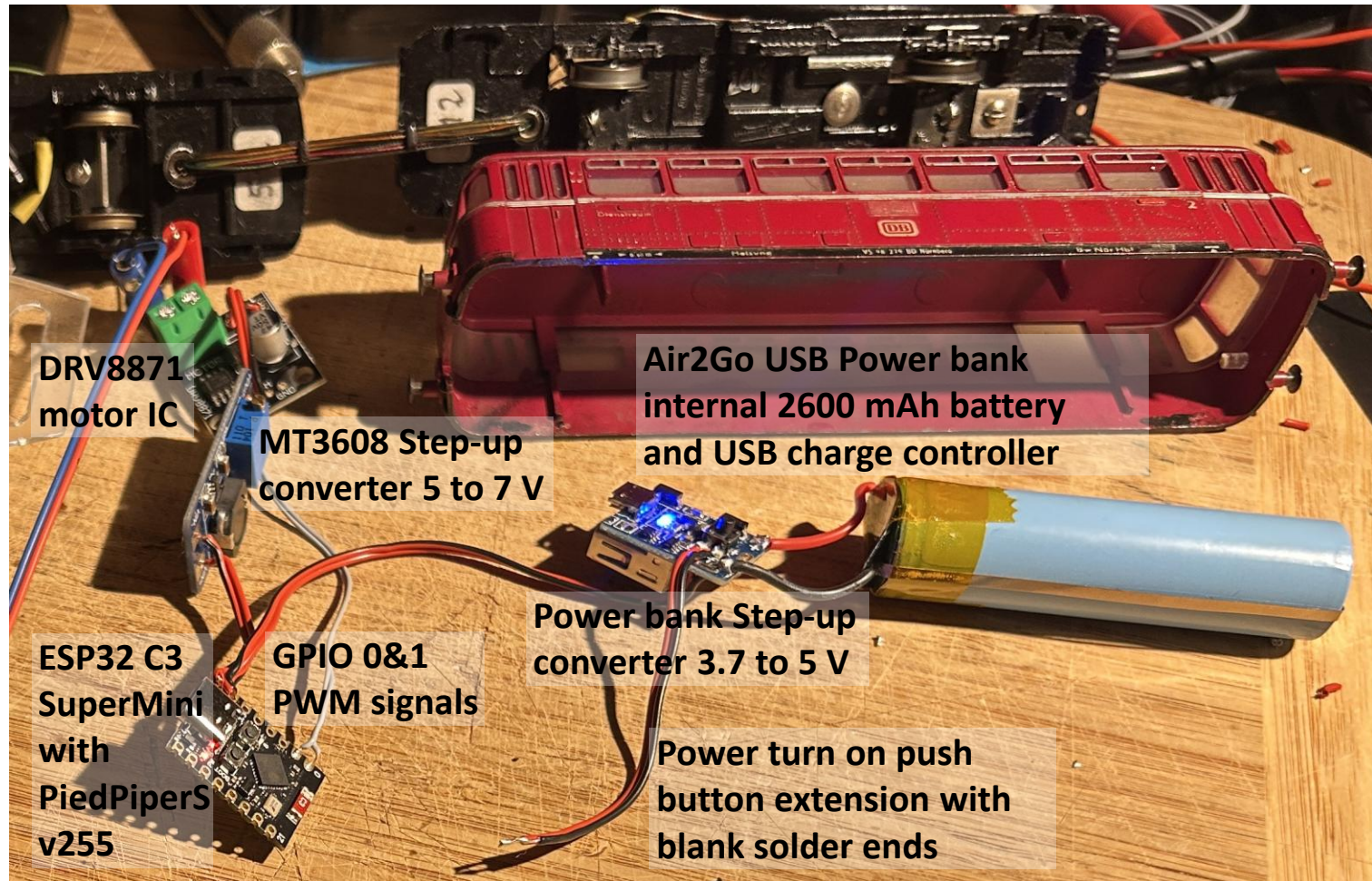
Touch code	Command for speed level	Meaning
•	--	decrease speed
••	-----	slow down
•••	-----	brake to halt
••••	0	fast brake and stop
•••••	00<	fast stop, reverse direction
• —	++	increase speed
• — —	+++++++	speed up
• — — —	+++++++	go fast
—	0	fast brake and stop
— —	00	fast brake and stop
————	00	fast brake and stop
— — —	?	info?

PiedPiper LED indicators

	LEDs  
Information on motor direction and speedlevel 0 to 32: • forward = 1 = blue LED + 0 to 16 short LED flashes • backward = 0 = red LED + 0 to 16 short LED flashes	 
Adjustment of motor speed: • 2 blue LED flash = speed level + 2; • 2 red LED flash = speed level - 2; • green & red LEDs long flash = fast brake, speed level = 0	  
Change of motor direction: • blue/red/blue + blue LED flash = forward = 1 • red/blue/red + red LED flash = backward = 0	 
Program running and main loop frequency: • orange flash after 5,000 main loop cycles, 200 times monitoring for input activity and up to 5 speed adjustments	

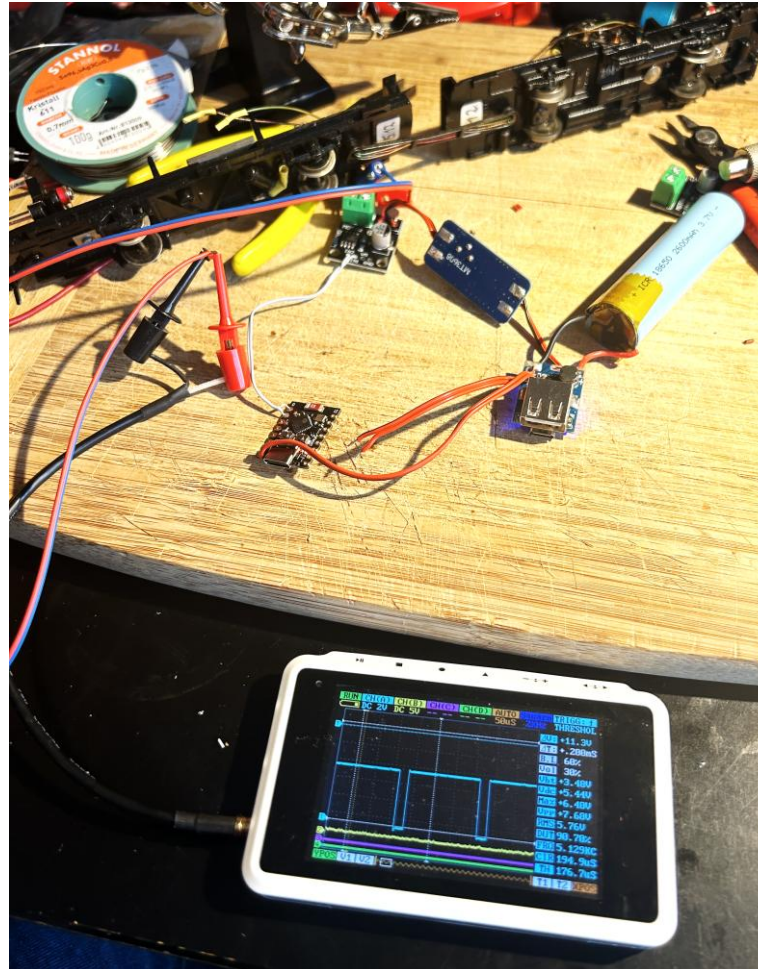
PiedPiperS integrated version

<https://github.com/jorail/PiedPiperS>



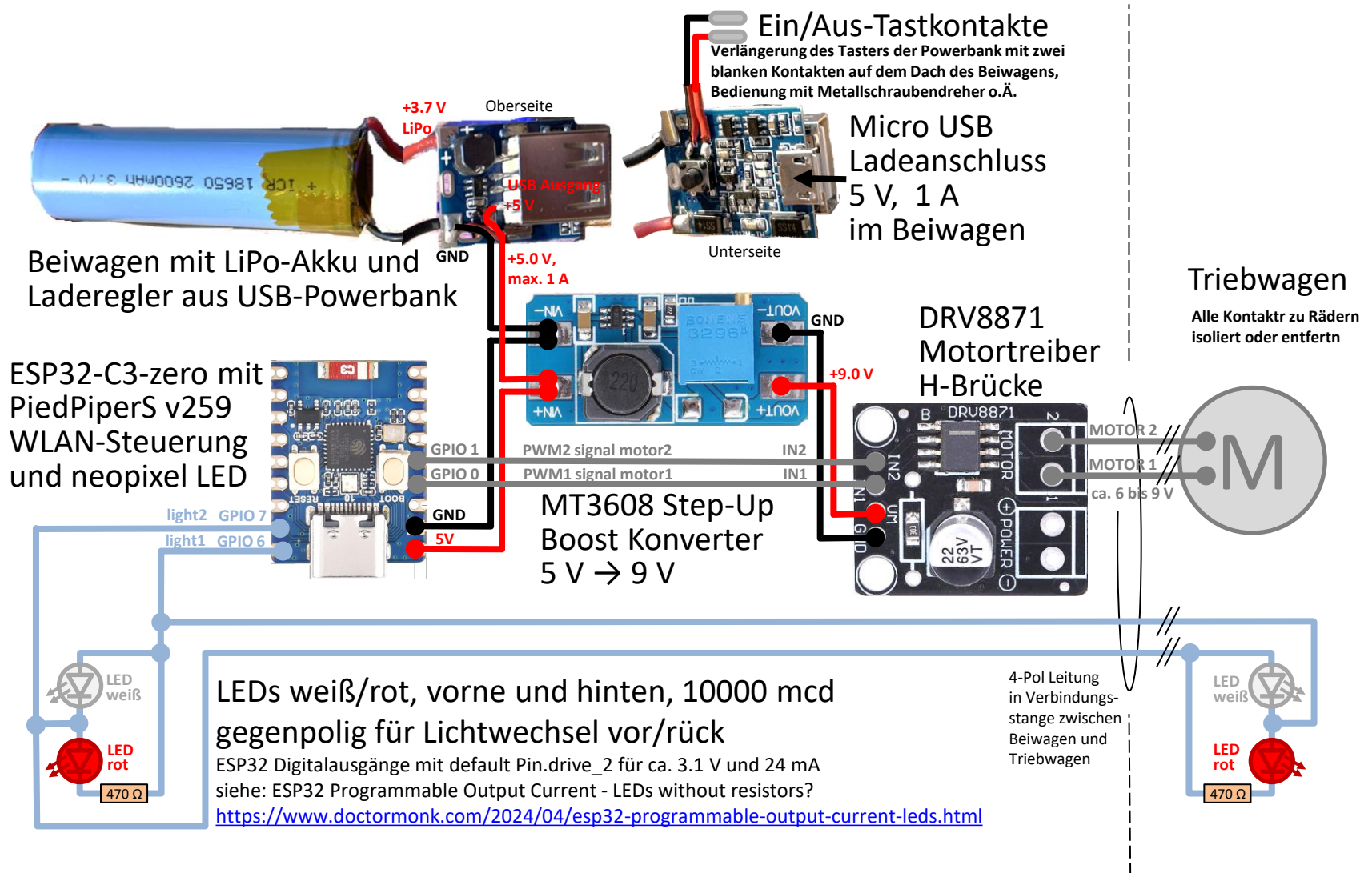
PiedPiperS integrated version

<https://github.com/jorail/PiedPiperS>



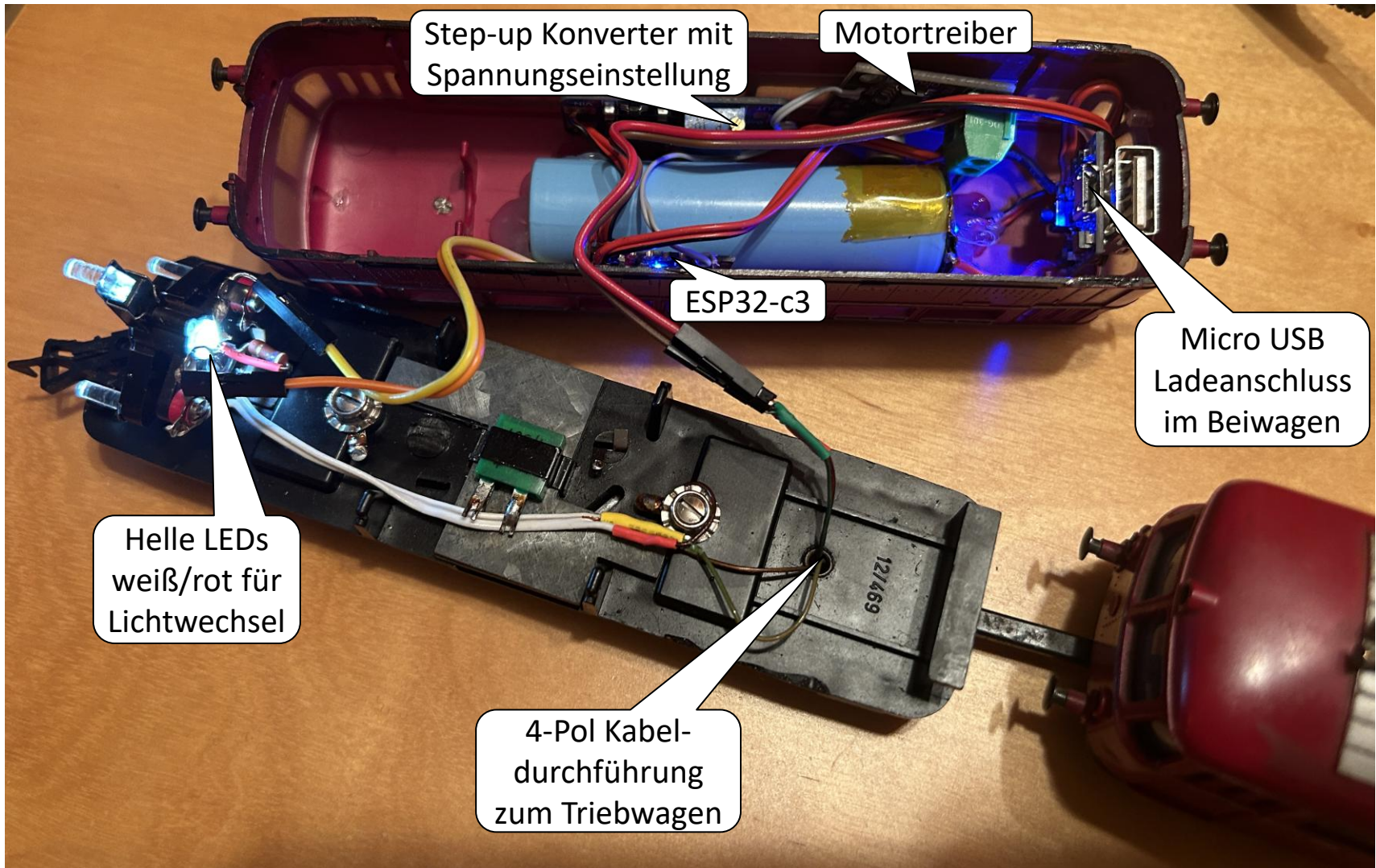
PWM gesteuerte
Ausgangsspannung
der H-Brücke für
Motorregelung

PiedPiperS v259 für Schienenbus VT 98



Schienenbus VT98 Beiwagen

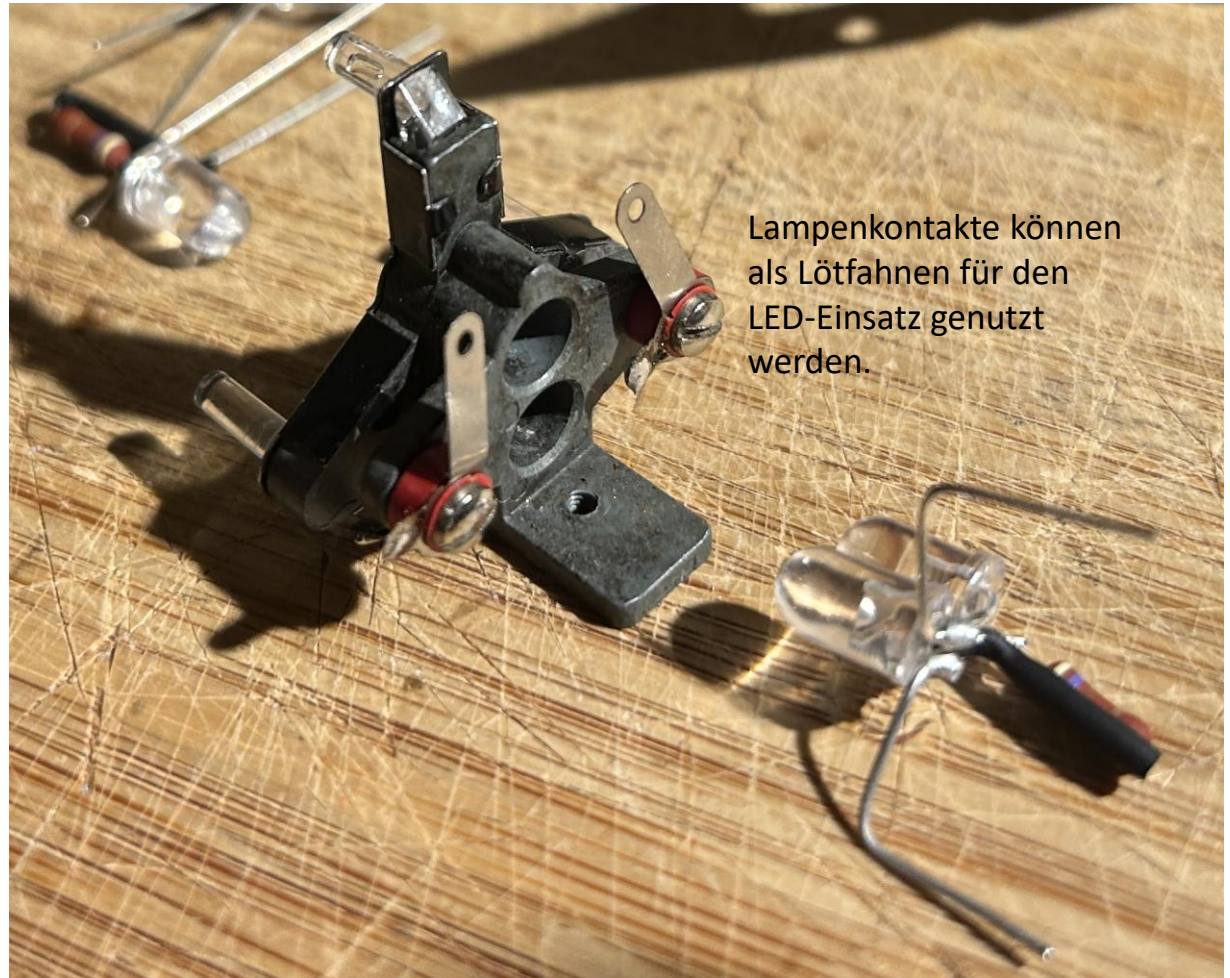
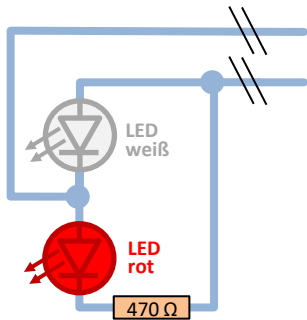
LiPo Akku und Steuerung eingebaut



Stirnlichter

LED weiß/rot gegenpolig und mit Vorwiderstand für rote LED

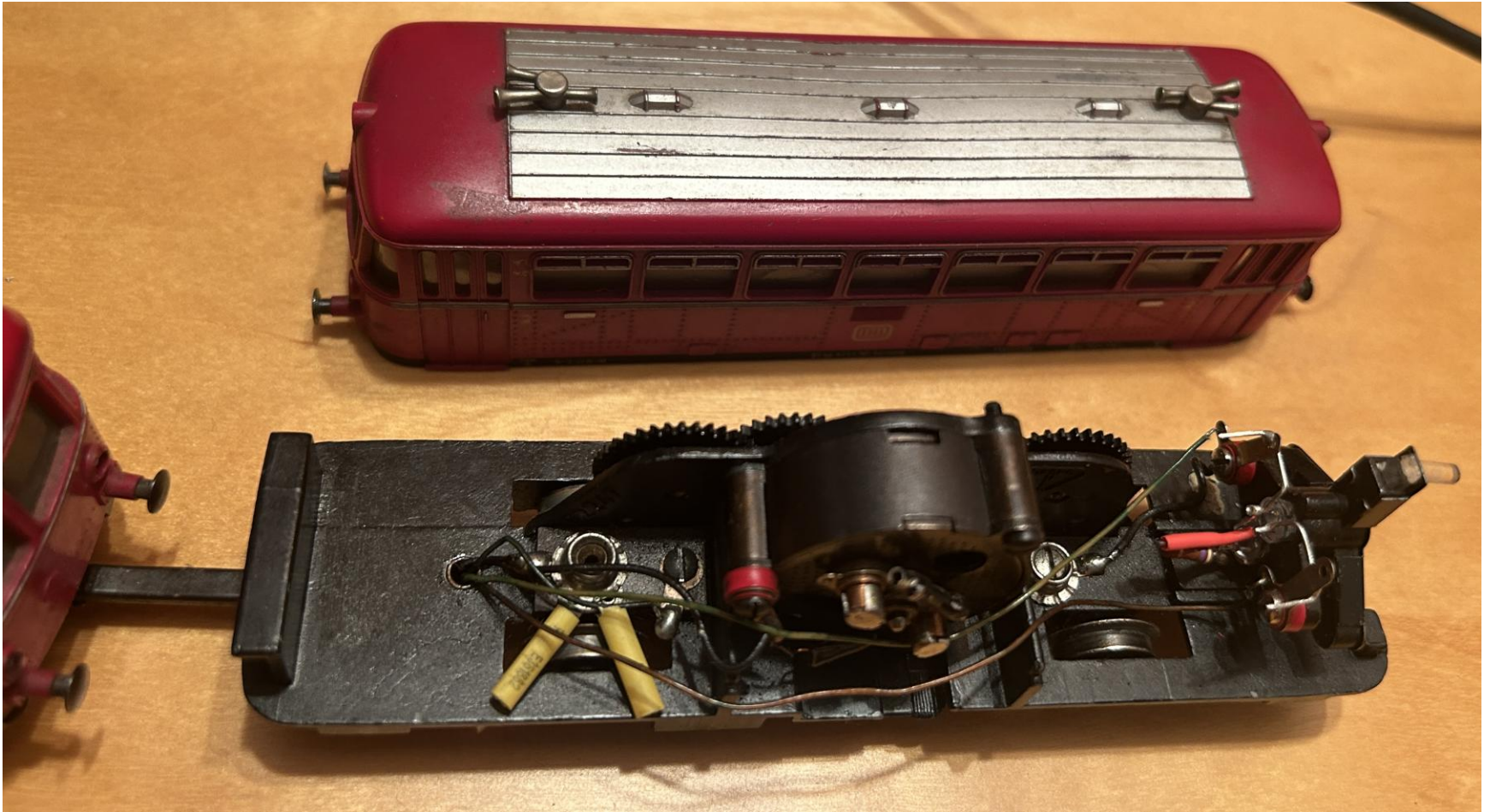
Vorwiderstand an LED-Typen anpassen: hier für ca. 3.1 V weiße LED 3,6 V ohne, und rote LED 2,2 V mit 470 Ω



Lampenkontakte können als Lötflächen für den LED-Einsatz genutzt werden.

Schienenbus VT98 Triebwagen

mit isolierten Radkontakten und LED-Beleuchtung



Schienenbus VT98 Beiwagen

An/Aus durch Tastkontakt auf dem Dach



Kompakte und simple
Bedienung An/Aus
von außen mit jedem
Strom leitenden
Gegenstand möglich

